

SOCIOL 723: Social Statistics II

Week 12, Machine Learning: Prediction and Regularization

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Today

The Bias–Variance Tradeoff

Regularization

Trees and Ensembles

A Different Goal

Weeks 5–11 asked: *what is the causal effect, and what must be true to recover it?*
Today we set that aside entirely.

Prediction versus inference

Inference: we care about a parameter β , its interpretation, and its standard error.

Prediction: we care only about $\hat{Y}$ for units we have not seen. The parameters are means, not ends, and may not even be identified.

- These goals conflict. The model that predicts best is usually not the model whose coefficients you would want to interpret, and vice versa.
- Molina and Garip (2019): sociology has plenty of genuine *prediction* problems (who drops out, which neighbourhoods gentrify, which cases to audit), and separate ML tools for describing heterogeneity.
- Next week we put the two back together. Today, keep them apart.

Bias-Variance

Decomposing Prediction Error (cont.)

Let $Y = f(X) + \epsilon$ with $E[\epsilon | X] = 0$ and $\text{Var}(\epsilon | X) = \sigma^2$, and let $\hat{f}$ be estimated from a training sample drawn independently of the new observation. At a new point x_0 , taking expectations over *both* the training sample and the new outcome:

$$E\left[(Y_0 - \hat{f}(x_0))^2\right] = \underbrace{\left(E[\hat{f}(x_0)] - f(x_0)\right)^2}_{\text{bias}^2} + \underbrace{\text{Var}(\hat{f}(x_0))}_{\text{variance}} + \underbrace{\sigma^2}_{\text{irreducible}}.$$

- **Bias:** how far the method is wrong on average, across training samples. Rigid models (a straight line) have high bias.
- **Variance:** how much $\hat{f}$ moves if you redraw the training data. Flexible models (a deep tree) have high variance.
- **Irreducible error:** noise. No method beats it.

Decomposing Prediction Error (cont.)

- Every tuning parameter in this course is a dial between the first two terms. Penalty size, tree depth, number of neighbours, bandwidth (Week 9!), all the same dial.

Why Unbiasedness Stops Being the Goal

Weeks 1–2 prized unbiasedness: Gauss–Markov ranked estimators *within* the unbiased class.

For prediction, that is the wrong criterion

We care about total error. Accepting a little bias to remove a lot of variance lowers MSE. Every method today is deliberately biased.

- OLS with k regressors has variance growing in k : with p/n not small, the variance term swamps everything and OLS predicts badly even though it is unbiased.
- When $p > n$, OLS is not even defined, $\mathbf{X}'\mathbf{X}$ is singular.
- **In-sample fit is not evidence.** R^2 never falls when you add regressors, so training error is a systematically optimistic estimate of test error. The gap is the *optimism*, and it grows with model complexity.

Estimating Test Error: Cross-Validation (cont.)

Split the data, fit on part, evaluate on the rest.

K -fold cross-validation

Partition the sample into K folds. For $k = 1, \dots, K$: fit on all folds but k , predict fold k , record the error. Average:

$$CV_K = \frac{1}{K} \sum_{k=1}^K \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \left(Y_i - \hat{f}^{(-k)}(X_i) \right)^2.$$

- $K = 5$ or 10 is standard: leave-one-out ($K = n$) has low bias but high variance and is expensive.

Estimating Test Error: Cross-Validation (cont.)

- Everything data-dependent must happen inside the fold, variable selection, standardization, imputation. Selecting predictors on the full sample and then cross-validating is a very common and very serious leak.
- With dependence (panels, clusters, time) split on the independent unit, not on rows. Same rule as the bootstrap in Week 2.

Regularization

Ridge Regression (cont.)

Add a penalty on the squared size of the coefficients:

$$\hat{\boldsymbol{\beta}}^{\text{ridge}} = \arg \min_{\boldsymbol{\beta}} \sum_{i=1}^n (Y_i - \mathbf{X}'_i \boldsymbol{\beta})^2 + \lambda \sum_{j=1}^p \beta_j^2, \quad \hat{\boldsymbol{\beta}}^{\text{ridge}} = (\mathbf{X}'\mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}'\mathbf{y},$$

where, as always with a penalty, $\mathbf{X}$ and $\mathbf{y}$ are centred and the intercept is left out of both the penalty and the displayed matrix.

- Adding $\lambda \mathbf{I}$ makes the matrix invertible **even when $p > n$** , which is why ridge works where OLS does not exist.
- Coefficients shrink toward zero but never reach it: ridge does not select.
- **Always standardize the predictors first**; the penalty is not scale-invariant, and the intercept is not penalized.

Ridge Regression (cont.)

- Ridge shines under **dense** signals: many small effects, and correlated predictors whose coefficients OLS cannot separate.
- As $\lambda \rightarrow 0$ we recover OLS; as $\lambda \rightarrow \infty$ all slopes go to zero and the fit collapses to the sample mean of Y . The dial is the bias–variance dial.

Lasso

Penalize the *absolute* size instead:

$$\hat{\beta}^{\text{lasso}} = \arg \min_{\beta} \sum_{i=1}^n (Y_i - X_i' \beta)^2 + \lambda \sum_{j=1}^p |\beta_j|.$$

- The ℓ_1 penalty sets coefficients exactly to zero, so lasso performs variable selection and estimation simultaneously.
- Geometrically: the ℓ_1 constraint region is a diamond with corners on the axes, and the contours of the sum of squares typically first touch it at a corner. The ℓ_2 region is a ball with no corners, hence no exact zeros.
- Lasso suits **approximate sparsity**: a few predictors matter, the rest are approximately irrelevant.
- **Elastic net** mixes them, $\lambda(\alpha \sum |\beta_j| + \frac{1-\alpha}{2} \sum \beta_j^2)$, which handles groups of correlated predictors better than pure lasso.

Choosing λ , and a Warning About Interpretation

- **Cross-validation** over a grid of λ : pick the minimizer, or the “one standard error” rule, the largest λ within one standard error of the minimum, which buys a simpler model at almost no predictive cost.
- There are also **theoretically derived** penalties that set λ from the model rather than by cross-validation. They matter when the lasso is a *step inside* an inference procedure rather than the final answer.

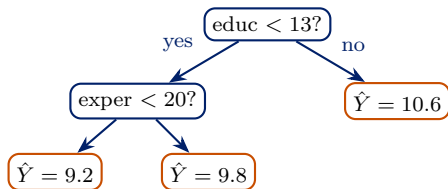
Lasso selection is not a variable-importance ranking

Among correlated predictors, lasso picks one essentially arbitrarily and zeroes the rest. Re-run on a bootstrap sample and you may get a different set. **Do not interpret the selected set as “the variables that matter”**, and do not report post-lasso p -values from a naive refit, the selection invalidates them. Valid post-selection inference is exactly what Week 13 is for.

Trees

Regression Trees (cont.)

Recursively split the covariate space to minimize within-node sum of squares; the prediction in a leaf is the mean of the training outcomes there.



- **Strengths:** automatic interactions and nonlinearity; invariant to monotone transformations; handles mixed variable types; readable.
- **Weaknesses:** high variance, a small change in the data can change the first split and hence everything below it. A single deep tree overfits badly; a shallow one underfits.

Regression Trees (cont.)

- Pruning with a complexity penalty helps but does not solve it. The real fix is to average many trees.

Bagging, Random Forests, Boosting

- **Bagging:** fit trees on bootstrap resamples and average. Averaging B nearly independent estimates cuts variance by roughly $1/B$ without adding bias. But bootstrap trees are *correlated*, they all split on the same dominant predictor first, which caps the gain.
- **Random forests:** bag, but at each split consider only a random subset of m predictors. Decorrelating the trees is what makes the averaging work. m is the main tuning parameter, alongside node size. Out-of-bag error is a free cross-validation estimate.
- **Boosting:** fit trees *sequentially*, each on the residuals of the current ensemble, adding a shrunk version: $\hat{f} \leftarrow \hat{f} + \eta \hat{t}$. Attacks *bias* rather than variance, and with small η and many trees is usually the strongest tabular predictor available.

Bagging and random forests reduce variance; boosting reduces bias. That is the cleanest way to keep them straight.

What ML Buys, and What It Does Not (cont.)

Buys

- Prediction with many predictors, including $p > n$.
- Automatic nonlinearity and interactions, no functional form to guess.
- Honest out-of-sample performance estimates.

Does not buy

- **Causal interpretation.** A variable importance score is not an effect.
- Valid standard errors for the coefficients it selects.
- Extrapolation: trees do return a prediction for out-of-range inputs, but only the boundary leaf's constant, so they cannot extend a trend.
- Freedom from the identification problems of Weeks 5–11.

What ML Buys, and What It Does Not (cont.)

The genuine sociological uses

Prediction as the goal (screening, triage, forecasting); *measurement* (constructing variables from text or images); *describing heterogeneity*; and **estimating nuisance functions inside a causal estimator**, which is next week.

Looking Ahead

- Every method today trades bias for variance, tuned by cross-validation.
- That trade is exactly why we cannot drop these estimates into a regression and read off a coefficient: they are *deliberately biased* and converge slowly.
- **Next week:** what has to change before an estimate this biased can be used as an ingredient in a causal estimator, and what it buys us when it can.

Reading

ISL Chapters 5, 6, and 8; Molina and Garip (2019). Optional: ISL Chapter 10 (deep learning); Kleinberg et al. (2015) on prediction policy problems.

References

- James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani. 2021. *An Introduction to Statistical Learning*, 2nd ed. Springer. [Ch. 5, 6, 8]
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